Curriculum Vitae

Rishabh Gupta

M.Sc. Physics, Indian Institute of Technology Indore

Indore, Madhya Pradesh, India

rishabh200207@gmail.com   +91-9903221158

ORCID   INSPIRE-HEP   LinkedIn   GitHub

Education

Jul 2024 – Jun 2026 (Exp.)

M.Sc. in Physics
Indian Institute of Technology Indore, India

- Thesis title: *Exploring identified hadron production at the LHC with machine learning and MAGE-HEP for reproducible Monte Carlo workflow*
  Thesis Supervisor: *Prof. Raghunath Sahoo*
- Cumulative Performance Index (CPI): 9.45 / 10

Sep 2021 – Jun 2024

B.Sc. (Honours) in Physics
Adamas University, West Bengal, India

- Dissertation: *Optimization of a THGEM-Based Detector for Cosmic Muons*
  Dissertation Supervisor: *Dr. Purba Bhattacharya*
- CGPA: 8.68 / 10

Experimental / Detector Experience

- Experience with gaseous detector systems (THGEM) including simulation and response studies
- Exposure to RPC based detectors for muon tracking in tomography applications

Publications

Bhattacharya, P., **Gupta, R.**, Guha, S., Rout, P. K., Datta, J., Majumdar, N., & Mukhopadhyay, S. (2025). *Study of charge dynamics in THGEM-based detectors – A numerical approach.* Journal of Instrumentation (JINST), **20**(04), C04019. doi:10.1088/1748-0221/20/04/C04019

Gupta, R., Guha, S., Bhattacharya, P., Mukhopadhyay, S., & Majumdar, N. *Python automation of THGEM simulation using COMSOL.* Proceedings of the National Conference on Frontiers in Modern Physics (NCFMP 2024), Springer. *(Accepted)*

Gupta, R., Goswami, K., Prasad, S., & Sahoo, R. (2026). *Inferring identified hadron production in pp collisions with physics-informed machine learning at the LHC.* arXiv preprint, submitted to PRX Intelligence, arXiv:2605.09022 [hep-ph].

Gupta, R., Goswami, K., Prasad, S., & Sahoo, R. (2026). *MAGE-HEP: Monte Carlo Analysis and Graphical Environment for High-Energy Physics.* arXiv preprint, arXiv:2605.17871 [hep-ph].

Conferences and Presentations

2024	Poster Presentation, *8th International Conference on Micro-Pattern Gaseous Detectors*, University of Science and Technology of China (USTC), Hefei, China
2023	Paper Presentation, *10th International Conference on Microelectronics, Circuits and Systems*, India

Research Experience

Aug 2025 – Present — **Master's Project**, Indian Institute of Technology Indore

- Developed staged training methodology for physics-informed neural networks on particle spectra (pp, $\sqrt{s} = 13.6$ TeV) with ratio and smoothness loss functions, validated against PYTHIA8. Extended to zero-shot rare particle detection using partial training with branching ratio constraints.
- MAGE: C++17 framework for reproducible Monte Carlo workflows with UUID tracking and automated ROOT output.

Jun 2025 – Present — **Continuing Research**, Saha Institute of Nuclear Physics, Kolkata

- Extension to 3D point cloud–based object detection for muon tomography
- Volumetric reconstruction and localisation studies

Dec 2024 – Feb 2025 — **Research Intern**, Saha Institute of Nuclear Physics, Kolkata

- CNN-based deep learning model for muon tomography
- Validation using simulated and reconstructed muon data

Jan 2024 – Jun 2024 — **Undergraduate Research**, Adamas University

- Optimisation of THGEM-based detectors for cosmic muons
- Detector simulations and efficiency studies

Jun 2023 – Aug 2023 — **Research Intern**, IISER Kolkata

- Polarimetry, Mueller matrix and Jones calculus

Software & Pipelines

Collider Particle Dynamics (CPD) *Author & Developer*

Feb 2024 – Sep 2024

Developed a framework to automate COMSOL based simulations of gaseous detectors, allowing efficient generation of model variations, parameter scans, and export of simulation results.

- Designed a command based workflow to generate hierarchical detector model configurations.
- Automated geometry edits, physics setup, and batch execution of simulation runs.
- Built utilities for logging simulations, exporting data tables, and generating plots.
- Used in detector simulation studies reported in *Journal of Instrumentation (JINST)*.
  doi: 10.1088/1748-0221/20/04/C04019

MAGE (Monte Carlo Assisted Generation Environment) *Author & Developer*

Jan 2026 – Present

Modular C++17 Qt-based IDE for reproducible Monte Carlo workflows. Node based context interface for event generation pipelines with multiple backends.

- Node-based workflow design: Connect generators, filters, and analysis modules
- PYTHIA8 backend fully implemented; AMPT integration in progress
- UUID-based run tracking and automated ROOT file generation for reproducible analysis
- Physics observable abstraction layer for systematic parameter variation
- Used in master's research for p_T spectra modelling across kinematic regions
- Designed as general-purpose HEP infrastructure; planned open-source release post-v2

Technical Skills

Programming	C++, Python
Simulation	GEANT4, COMSOL Multiphysics, ROOT
MC Tools	PYTHIA8, AMPT, EPOS
Hardware	Detector characterisation, silicon detectors (theory), RPC/scintillator systems
ML/DL	PyTorch, Hyperparameter Optimization, physics-informed loss design, flow models, Gen AI
Typesetting	LaTeX

Awards and Academic Achievements

2024	IIT–JAM Physics, All India Rank 435
2020	NASA Ames Space Settlement Contest, 3rd Prize
2017	NASA Ames Space Settlement Contest, Honourable Mention

Languages

English, **Hindi**, **Bengali**